

Project Scope: **lecture 1-2**

Maximum Grade: 5

Student Name:

Student ID:

Assignment Rules & Regulations:

- Plagiarism, cheating, or any form of academic dishonesty is **strictly prohibited**. All work submitted must be the student's own, and all external sources, such as code snippets, libraries, or academic papers, must be properly cited.
- Requests for extensions must be requested **in advance of the deadline** and should be supported by valid and documented reasons, such as medical or family emergencies. Extensions will be granted at the instructor's discretion.
- If you encounter any issues or require clarifications related to the assignment, promptly communicate with the instructor or teaching assistant. **Do not share your assignment solutions with classmates before the submission deadline.**

Mini-Project Objective:

- **Data organization**
- **Data presentation**
- **Data summarization I: Measure of central tendency**
- **Data summarization II: Measure of variation**

Write a python code to construct 8 classes frequency distribution table that contains the following columns (**class limits, class boundaries, class frequency, class relative frequency, accumulated frequency and accumulated relative frequency**) for the given data that represents a *sample* of the achievement test scores of 100 students applying for a master's degree in SUT. Import the scores from the given text file named *data1.txt*. Then use the output to draw *histogram, frequency polygon, and ogive plot* for the class frequencies. Also, calculate the *mean, standard deviation, and coefficient of variation* for both raw data and grouped data.

Deliverables (individual submission):

1. The python code for data importing, constructing the frequency distribution table, plotting the required diagrams, and calculating the required statistics.
2. A technical report illustrating the code, frequency distribution table, graphs and different statistics.
3. A soft copy and a hard copy of each of the above requirements have to be submitted.

Deadline for Submission and Discussion: Sunday, December 14th, 2025.

